

Chapitre 2

Configuring Core Network Services

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Plan

- Planning your IP address layout
- Installing and configuring a DHCP server
- Installing and configuring a DNS server

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Planning your IP address layout

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IP address layout

- What types of machines (e.g., servers, PCs) you'll need to provide addresses to.
- Having multiple subnets would certainly help. With **subnetting**, you can create individual networks for each type of service (**not covered in this module**).
- When **planning an IP addressing scheme**, you constantly ask questions like:
 - How many usable IP addresses do I get?
 - Where is the network address?
 - What is the broadcast address?
 - Is this subnet too big or too small?
- **ipcalc** answers all of these instantly, before you configure anything.
 - Installing ipcalc
 - ✓ `sudo apt-get update`
 - ✓ `sudo apt-get install ipcalc`

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IP address layout

- Basic usage for testing a network idea
 - Suppose you are *thinking* about using this subnet: 10.0.70.0/24
 - Before committing to it, run: **ipcalc 10.0.70.0/24**

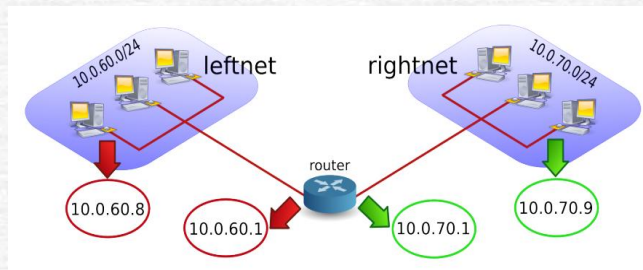
```
bakhouya@ubuntu:~$ sudo ip netns exec R ipcalc 10.0.70.0/24
Address: 10.0.70.0      00001010.00000000.01000110. 00000000
Netmask: 255.255.255.0 = 24  11111111.11111111.11111111. 00000000
Wildcard: 0.0.0.255      00000000.00000000.00000000. 11111111
=>
Network: 10.0.70.0/24    00001010.00000000.01000110. 00000000
HostMin: 10.0.70.1      00001010.00000000.01000110. 00000001
HostMax: 10.0.70.254    00001010.00000000.01000110. 11111110
Broadcast: 10.0.70.255  00001010.00000000.01000110. 11111111
Hosts/Net: 254          Class A, Private Internet
```

- 10.0.70.0: Cannot be assigned to a host
- 10.0.70.255: Used to talk to all hosts, not assignable
- 10.0.70.1 – 10.0.70.254; HostMin / HostMax, these are the usable Ips
- 254: Hosts/Net, maximum number of devices supported

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IP address layout

- For this example, we have:
 - 1 router interface
 - 6 hosts
- That's 3 IPs minimum.
- With ipcalc, we immediately see:
 - /24 gives 254 hosts → too much but simple
 - We could use /29, what do you think!



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IP address layout

- Important to plan ahead IP address layout.
 - Plan your IP addressing before building anything.
 - Always split the network into **subnets**.
 - Plan **bigger** than your current needs.
 - Choose IP ranges that allow **growth**.
 - Use **tools** to validate your plan.

When you develop your project using Linux namespaces, you **must follow these IP planning practices**

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Picturing DHCP

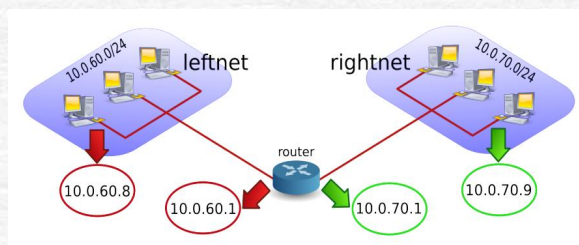
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Installing and configuring a DHCP server

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DHCP server

- Set up a DHCP server and configure it to serve IPv4 addresses to our network.
 - Use **Linux network namespaces**, not virtual machines.
 - All namespaces share the same filesystem.
 - DHCP is installed once on the host.
 - Configuration and execution happen inside the namespace.
 - DHCP server runs on the router namespace R.



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DHCP server

- Installing DHCP server in the router
 - `sudo ip netns exec $NS apt update`
 - `sudo ip netns exec $NS apt install -y isc-dhcp-server`
- Tell DHCP which interfaces to serve


```
sudo ip netns exec R bash -c 'cat > /etc/default/isc-dhcp-server <<EOF
INTERFACESv4="Rr RI"
EOF'
```

```
# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
#       Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4=""
INTERFACESv6=""
bakhouya@ubuntu:~/Documents/Cours/4-DHCP$
```
- DHCP server global options


```
sudo ip netns exec R bash -c 'cat > /etc/dhcp/dhcpd.conf <<EOF
default-lease-time 600; # DHCP leases last 600 seconds (10 minutes) by default.
max-lease-time 7200; #Maximum lease time is 2 hours.
authoritative; # This server is the main authority for these subnets. Prevents
clients from using stale IP info from other DHCP servers.
```

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DHCP server

- Right LAN DHCP configuration


```
subnet 10.0.60.0 netmask 255.255.255.0 { # defines the Right LAN subnet
    range 10.0.60.100 10.0.60.150; # DHCP assigns IPs only in this range.
    option routers 10.0.60.1; # default gateway for clients in Right LAN.
    option subnet-mask 255.255.255.0; # standard network info
    option broadcast-address 10.0.60.255; # standard network info
    option domain-name-servers 8.8.8.8; # tells clients to use Google DNS.
}
```
- Left LAN DHCP configuration


```
subnet 10.0.70.0 netmask 255.255.255.0 {
    range 10.0.70.100 10.0.70.150;
    option routers 10.0.70.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 10.0.70.255;
    option domain-name-servers 8.8.8.8;
}
EOF'
```
- DHCP server can now serve **two isolated LANs** at the same time.

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DHCP server

- Check the files /etc/default/isc-dhcp-server and /etc/dhcp/dhcpd.conf

```
bakhouya@ubuntu:~/Documents/Cours/4-DHCP$ sudo ip netns exec R cat /etc/default/isc-dhcp-server
INTERFACESv4="Rr Rl"
bakhouya@ubuntu:~/Documents/Cours/4-DHCP$ sudo ip netns exec R cat /etc/dhcp/dhcpd.conf
default-lease-time 600;
max-lease-time 7200;
authoritative;

# Right LAN
subnet 10.0.60.0 netmask 255.255.255.0 {
    range 10.0.60.100 10.0.60.150;
    option routers 10.0.60.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 10.0.60.255;
    option domain-name-servers 8.8.8.8;
}

# Left LAN
subnet 10.0.70.0 netmask 255.255.255.0 {
    range 10.0.70.100 10.0.70.150;
    option routers 10.0.70.1;
    option subnet-mask 255.255.255.0;
    option broadcast-address 10.0.70.255;
    option domain-name-servers 8.8.8.8;
}
bakhouya@ubuntu:~/Documents/Cours/4-DHCP$ cat
```

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DHCP server

- Start DHCP server
 - sudo ip netns exec R systemctl restart isc-dhcp-server
 - sudo ip netns exec R systemctl enable isc-dhcp-server
- Check status
 - Shows if the DHCP server is running correctly inside namespace R.
 - sudo ip netns exec \$NS pgrep -a dhcpd || echo "DHCP server is not running!"

```
bakhouya@ubuntu:~/Documents/Cours/4-DHCP$ sudo ip netns exec R pgrep -a dhcpd || echo "DHCP server is not running!"
446404 dhcpd -cf /etc/dhcp/dhcpd.conf Rr Rl
```

- Test DHCP clients (Cl and Cr)
 - sudo ip netns exec Cl dhclient -v veth-Cl || echo "DHCP failed for Cl"
 - sudo ip netns exec Cr dhclient -v veth-Cr || echo "DHCP failed for Cr"
- Show assigned IPs
 - sudo ip netns exec Cl ip addr show veth-Cl
 - sudo ip netns exec Cr ip addr show veth-Cr

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DHCP server

- Assigned DHCP IPs on client interfaces (4_enabling_dhcp)

```

Cl assigned IP:
110: veth-Cl@if109: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen
1000
    link/ether 66:5c:db:4b:8c:0c brd ff:ff:ff:ff:ff:ff link-netns R
    inet 10.0.70.100/24 brd 10.0.70.255 scope global dynamic veth-Cl
        valid_lft 516sec preferred_lft 516sec
Cr assigned IP:
112: veth-Cr@if111: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen
1000
    link/ether aa:c4:65:3f:12:2c brd ff:ff:ff:ff:ff:ff link-netns R
    inet 10.0.60.100/24 brd 10.0.60.255 scope global dynamic veth-Cr
        valid_lft 518sec preferred_lft 518sec
=== DHCP server setup and testing completed successfully ===
bakhouya@ubuntu:~/Documents/Cours/4-DHCP$ sudo ip netns exec Cl ping -c 3 10.0.60.100
PING 10.0.60.100 (10.0.60.100) 56(84) bytes of data.
64 bytes from 10.0.60.100: icmp_seq=1 ttl=63 time=0.111 ms
^C
--- 10.0.60.100 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.111/0.111/0.111/0.000 ms
bakhouya@ubuntu:~/Documents/Cours/4-DHCP$ sudo ip netns exec Cr ping -c 3 10.0.70.100
PING 10.0.70.100 (10.0.70.100) 56(84) bytes of data.
64 bytes from 10.0.70.100: icmp_seq=1 ttl=63 time=0.056 ms
^C

```

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DHCP server

- Test connectivity
 - sudo ip netns exec Cl ping -c 3 10.0.60.8
 - sudo ip netns exec Cr ping -c 3 10.0.70.9
- Release the DHCP lease
 - sudo ip netns exec Cl dhclient -r veth-Cl
 - sudo ip netns exec Cr dhclient -r veth-Cr
- Flush IP addresses from the interface
 - sudo ip netns exec Cl ip addr flush dev veth-Cl
 - sudo ip netns exec Cr ip addr flush dev veth-Cr

```

bakhouya@ubuntu:~/Documents/Cours/4-DHCP$ sudo ip netns exec Cl dhclient -v veth-Cl || echo "DHCP failed
for Cl"
Internet Systems Consortium DHCP Client 4.4.1
Copyright 2004-2018 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/veth-Cl/66:5c:db:4b:8c:0c
Sending on LPF/veth-Cl/66:5c:db:4b:8c:0c
Sending on Socket/fallback
DHCPDISCOVER on veth-Cl to 255.255.255.255 port 67 interval 3 (xid=0x1989ed18)
DHCPOFFER of 10.0.70.100 from 10.0.70.1
DHCPREQUEST for 10.0.70.100 on veth-Cl to 255.255.255.255 port 67 (xid=0x18ed8919)
DHCPACK of 10.0.70.100 from 10.0.70.1 (xid=0x1989ed18)
bound to 10.0.70.100 -- renewal in 260 seconds.
bakhouya@ubuntu:~/Documents/Cours/4-DHCP$ sudo ip netns exec Cl dhclient -r veth-Cl
Killed old client process
bakhouya@ubuntu:~/Documents/Cours/4-DHCP$ sudo ip netns exec Cl ip link show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
110: veth-Cl@if109: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP mode DEFAULT group
default qlen 1000
    link/ether 66:5c:db:4b:8c:0c brd ff:ff:ff:ff:ff:ff link-netns R

```

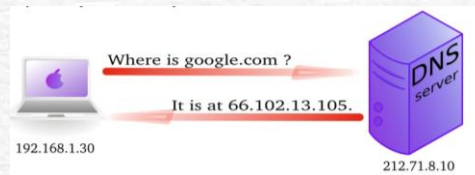
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Installing and configuring a DNS server

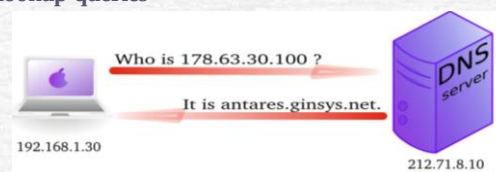
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Introduction to DNS

- Used by many network services to **translate names into network addresses** and to locate services on the network (by name).
- Whenever you visit a web site, send an e-mail, log on to Active Directory, chat, or use VoIP, there will be one or (many) more queries to dns services.
 - **Forward lookup queries**



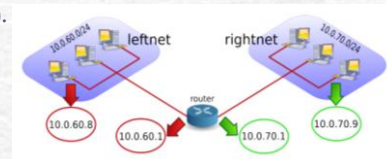
- **Reverse lookup queries**



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Introduction to DNS

- DNS makes navigating networked resources much easier.
- Unless you have a very small network, it's unlikely that you'll remember which IP addresses belong to which machines, e.g., it's easier to refer to the printer as hp-color-01 rather than remembering an IP address, such as 10.19.89.40 .
- DNS helps by mapping names to IP addresses, so you can refer to computers by their hostname and DNS will do the work of translating that back to the IP address.
- Creating a consistent and predictable naming scheme for all your networked resources is a great idea.
- Let us configure a DNS to resolve names inside the following network and not forward to the Internet (**Check Hands-on Labs**).

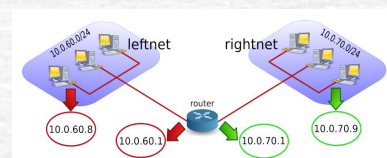


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Internal DNS server

- Create DNS namespace and veth connection to the router.
- Assign IP addresses
- Configure static routing (optional in small lab).

NS	IP	Role	Notes
Cl	10.0.70.9	Client	Default route → R
Cr	10.0.60.8	Client	Default route → R
R	10.0.70.1 / 10.0.60.1	Router	Forwards traffic
DNS	10.0.70.53	DNS server	Resolves internal names only



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Internal DNS server

- Adding a DNS Server in a Separate Subnet (**5_enabling_dns**).
 - Create a DNS server in its own subnet
 - Connect it to Router R
 - Allow C1 and C2 to reach the DNS server via routing only
 - **No DNS name translation configuration required so far.** Validate configuration (this MUST pass):
 - ✓ `sudo ip netns exec DNS named-checkconf`
 - ✓ `sudo ip netns exec DNS named-checkzone lab.test /etc/bind/db.lab.test`
 - ✓ `sudo ip netns exec DNS named-checkzone 30.168.192.in-addr.arpa /etc/bind/db.192.168.30`

```

=== Test 1: DNS process ===
[OK] DNS process (named) is running
=== Testing connectivity ===
Ping from C1 -> DNS:
PING 10.0.100.53 (10.0.100.53) 56(84) bytes of data.
64 bytes from 10.0.100.53: icmp_seq=1 ttl=63 time=0.045 ms

--- 10.0.100.53 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.045/0.045/0.045/0.000 ms
[OK] C1 can reach DNS
Ping from C2 -> DNS:
PING 10.0.100.53 (10.0.100.53) 56(84) bytes of data.
64 bytes from 10.0.100.53: icmp_seq=1 ttl=63 time=0.033 ms

--- 10.0.100.53 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.033/0.033/0.033/0.000 ms
[OK] C2 can reach DNS
=== DNS setup complete ===
DNS reachable at 10.0.100.53 from C1 and C2
bakhouya@ubuntu:~/Documents/Cours/4-DNS$

```

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Internal DNS server

- With this done, we actually have a working DNS server.
- Add records to it and build our configuration (**5_enabling_resolo**).
- Let's take a look at the default configuration folder: `/etc/bind`

```

bakhouya@ubuntu:~/Documents/Cours/4-DNS$ sudo ip netns exec DNS ls /etc/bind
db.10.0.100      db.lab.local    named.conf      named.conf.options
db.192.168.30   db.lab.test     named.conf.local

```

 - **named.conf** file: the main BIND configuration file that loads all other configuration files.
 - **named.conf.options**: defines global **DNS behavior**, e.g., as recursion and listening interfaces.
 - **named.conf.local**: declares the **DNS zones** handled by the server.
 - **db.lab.local** and **db.lab.test** files are forward lookup zone files that map **hostnames to IP addresses**.
 - **db.10.0.100** and **db.192.168.30** are **reverse lookup zone** files that map **IP addresses to hostnames**.
- **named.conf** is the main BIND configuration file that loads all other DNS configuration files.


```

bakhouya@ubuntu:~/Documents/Cours/4-DNS$ cat /etc/bind/named.conf
include "/etc/bind/named.conf.options";
include "/etc/bind/named.conf.local";

```

 - The root configuration file for BIND (named)
 - This is the first file loaded when DNS starts

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Internal DNS server

- **named.conf.options** is the global DNS behavior. It controls **global DNS server behavior** (how BIND listens, who can query it, recursion, forwarding, security).
 - Working directory for BIND
 - ✓ Where BIND stores: zone cache and temporary files.
 - ✓ You normally never touch this.
 - Allows recursive DNS queries
 - ✓ yes;: Clients (Cl / Cr) can ask “I don’t know the IP, go figure it out”
 - ✓ No: DNS would only answer authoritative zones

```
bakhouya@ubuntu:~/Documents/Cours/4-DNS$ sudo ip netns exec DNS cat /etc/bind/named.conf.options
options {
    directory "/var/cache/bind";
    recursion yes;
    allow-query { any; };
    listen-on { 10.0.100.53; };
    dnssec-validation no;

    forwarders {
        8.8.8.8;
        1.1.1.1;
    };
    forward only;
};
bakhouya@ubuntu:~/Documents/Cours/4-DNS$
```

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Internal DNS server

- Who is allowed to query this DNS
 - **Any**: all hosts can query .
 - Usually restricted to internal subnets.
- Which IP address BIND listens on (sudo ip netns exec DNS ss -ltnp | grep :53)
 - DNS listens only on 10.0.100.53
 - Not on 127.0.0.1
- Disables DNSSEC validation
 - In real-world setups, this is usually yes.
 - Avoids DNSSEC failures
- Upstream DNS servers (forwarders)
 - If DNS does NOT know the answer locally and **recursion** is enabled, then it forwards queries to Google DNS (8.8.8.8) and Cloudflare DNS (1.1.1.1).

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Internal DNS server

- **named.conf.local**: zone declarations
 - Declares which **DNS zones** this server is **authoritative** for.
 - ✓ "lab.local" is the **domain name** for your internal lab network. any query for ***.lab.local** (like cl.lab.local or cr.lab.local) is handled by this server.
 - ✓ **master** means this server is the **authoritative source** for this zone.
 - For the domain **lab.local**, look at /etc/bind/db.lab.local to answer queries."
 - Without this, queries like dig @10.0.100.53 cl.lab.local would fail—even if the DNS server is running.

```
bakhouya@ubuntu:~/Documents/Cours/4-DNS$ sudo ip netns exec DNS cat /etc/bind/named.conf.local
zone "lab.local" {
    type master;
    file "/etc/bind/db.lab.local";
};
```

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Internal DNS server

- **db.lab.local** : forward lookup zone (name → IP)
 - A forward zone file to convert hostnames into IP addresses

```
bakhouya@ubuntu:~/Documents/Cours/4-DNS$ sudo ip netns exec DNS cat /etc/bind/db.lab.local
$TTL 604800
@ IN SOA dns.lab.local. admin.lab.local. (
    20260207
    604800
    86400
    2419200
    604800 )
;
@ IN NS dns.lab.local.
dns IN A 10.0.100.53
cl IN A 10.0.70.9
cr IN A 10.0.60.8
```

@ → shorthand for the zone itself
(lab.local in this case).
IN → Internet class.
SOA → Start of Authority.

dns.lab.local. → Primary master name server for this zone.
admin.lab.local. → Email of the zone admin (replace @ with .).
20260207 → Serial number of the zone. Increment this when the file changes.
604800 → Refresh (secondary servers check for updates every 7 days).
86400 → Retry (if refresh fails, retry after 1 day).
2419200 → Expire (after 28 days, secondaries stop serving old data).
604800 → Negative TTL (how long to cache failed lookups).

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Internal DNS server

- **db.10.0.100** : Reverse lookup zone (IP → name)
 - A reverse DNS zone
 - Converts IP addresses into hostnames
 - **@** IN NS ns.lab.local: tells anyone querying the reverse zone that ns.lab.local. is authoritative for the reverse lookups of 10.0.100.x.
 - 53 IN PTR ns.lab.local
 - ✓ **53** → the last octet of the IP address **10.0.100.53**.
 - ✓ **PTR ns.lab.local.** → says that 10.0.100.53 maps back to ns.lab.local.

```
bakhouya@ubuntu:~/Documents/Cours/4-DNS$ sudo ip netns exec DNS cat /etc/bind/db.10.0.100
$TTL 604800
@ IN SOA ns.lab.local. admin.lab.local. (
    20260207
    604800
    86400
    2419200
    604800 )
;
@ IN NS ns.lab.local.
53 IN PTR ns.lab.local.
```

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Internal DNS server

- The DNS is already running with connectivity testing.
 - Creates forward and reverse zones for your lab.
 - Adds A records for cl, cr, and dns.
 - Adds reverse PTR records for the DNS IP.
 - Configures named.conf.local and named.conf.options with forwarding enabled.
 - Restarts BIND inside the DNS namespace and tests name resolution.
 - ✓ `sudo ip netns exec Cr ping -c 2 cl.lab.local`
 - ✓ `sudo ip netns exec Cl ping -c 2 cr.lab.local`
 - ✓ `sudo ip netns exec Cr dig cl.lab.local +short 10.0.70.9`
 - ✓ `sudo ip netns exec Cl dig cr.lab.local +short`

```
bakhouya@ubuntu:~/Documents/Cours/4-DNS$ sudo ip netns exec Cl dig @10.0.100.53 cl.lab.local A +short
10.0.70.9
bakhouya@ubuntu:~/Documents/Cours/4-DNS$ sudo ip netns exec Cr dig @10.0.100.53 cr.lab.local A +short
10.0.60.8
bakhouya@ubuntu:~/Documents/Cours/4-DNS$ sudo ip netns exec Cl ping -c 1 cl.lab.local
PING cl.lab.local (10.0.70.9) 56(84) bytes of data:
64 bytes from 10.0.70.9: icmp_seq=1 ttl=64 time=0.009 ms
```

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References

- Paul Cobbaut, Linux Networking, <http://linux-training.be/>
- Linux Network Administrator's Guide, 2nd Edition, by Olaf Kirch & Terry Dawson; published by O'Reilly & Associates, Inc. 503 pages; <https://www.linuxdoc.org/>
- Jay LaCroix, Mastering Linux Network Administration, Packt Publishing Ltd. ISBN 978-1-78439-959-7, www.packtpub.com
- Remo Suppi Boldrito, Network administration, <https://openaccess.uoc.edu/>